



Computer Networks

Week # 10

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URL- <https://sites.google.com/view/nadeemuolcscn/home/course-material>

ACKNOWLEDGMENTS

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Week 10: Course Plan

Network Layer

- Connection oriented vs connectionless service
- Logical Addressing (IP addressing)
 - Classful addressing

Network Layer

- Network layer is concerned with getting packets from the source **all the way** to the destination
 - May require many hops at intermediate routers (**multiple hops**),
 - rather than a single link, as in the data link layer (moving frames from one end of wire to the other)
- Its primary function is **routing**
- It deals with **end-end to transmission**
- It involves at the source host, destination host and all routers in the path

Connectionless vs. Connection Oriented

- Network layer should provide either connection oriented or connectionless service

Two major views are from ARPA Internet Community and Telecommunications community

- **ARPA Internet Community**
 - Routers' job is to move packets around (nothing else)
 - The network is inherently unreliable (no matter how it is designed)
 - leave error and flow control to the hosts (transport layer)
- Therefore, network service should be connectionless

Connectionless vs. Connection Oriented

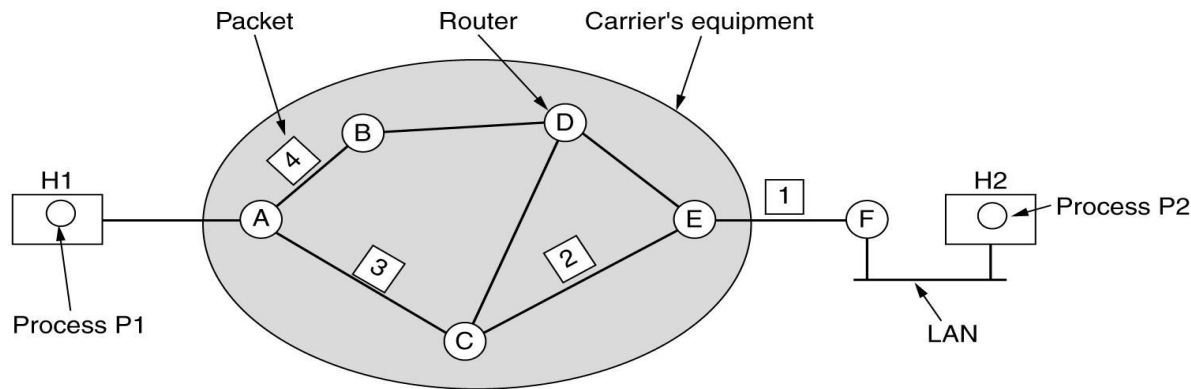
- **Telecommunications community (including ISO, ATM Forum, etc)**
 - network service should be (reasonably) reliable and connection-oriented
 - Their claim comes after 100 years of successful with the worldwide telephone system experience
 - QoS, a dominant factor, is very difficult to achieve without connections in the network, in particular, for real time traffic such as audio and video

Implementation of Connectionless Service

- In this service, packets are injected into the network directly and routed independently of each other
- No advance set up is needed
- So, packets are called **datagrams** and the network is called **datagram network**

Implementation of Connectionless Service

- Routing within a datagram subnet
- Due to traffic jam, packet 4 is sent on different route



The algorithm that manages tables and makes the routing decisions is called **routing algorithm**

A's table			
initially	later	C's table	E's table
A -	A -	A A	A C
B B	B B	B A	B D
C C	C C	C -	C C
D B	D B	D D	D D
E C	E B	E E	E -
F C	F B	F E	F F
Dest. Line			

Connection Oriented Service

- A path from the source router all the way to destination router must be established before any data packets can be sent
- This connection is called **Virtual Circuit (VC)** and the network is called **virtual-circuit network**

Connectionless VS. Connection-Oriented

Issue	Datagram subnet	Virtual-circuit subnet
Circuit setup	Not needed	Required
Addressing	Each packet contains the full source and destination address	Each packet contains a short VC number
State information	Routers do not hold state information about connections	Each VC requires router table space per connection
Routing	Each packet is routed independently	Route chosen when VC is set up; all packets follow it
Effect of router failures	None, except for packets lost during the crash	All VCs that passed through the failed router are terminated
Quality of service	Difficult	Easy if enough resources can be allocated in advance for each VC
Congestion control	Difficult	Easy if enough resources can be allocated in advance for each VC

Network Layer Core Protocols

Protocols which route data from a node or hop to another hop between two end hosts in a network are called **network-layer protocols**.

- **IP:** A connectionless unreliable protocol that is part of the TCP/IP protocol suite
- **ARP** (Address Resolution Protocol)
 - Resolves IP addresses to MAC addresses
- **ICMP** (Internet Control Message Protocol)
 - Diagnostics and error reporting
- **IGMP** (Internet Group Management Protocol)
 - Management of group multicast

Logical (IP) Addressing

Internet Assigned Number Authority (IANA)

- IANA oversees global IP addressing allocation
 - Responsible for global coordination
- IANA delegates Internet resources to the Regional Internet Registries (RIRs) who, in turn, follow their regional policies to delegate resources to their customers,
 - which include Internet Service Providers and end-user organizations.

Users are assigned IP addresses from ISPs

- ISPs obtain allocation of IP addresses from
 - Local Internet Registry (LIR),
 - National Internet Registry (NIR) or
 - Regional Internet Registry (RIR)

Local Internet Registry (LIR)

- An organization that has been allocated a block of IP addresses by a RIR, and that assigns most parts of this block to its own customers.
- Most LIRs are ISPs, enterprises, or academic institutions.
- Membership in an RIR is required to become an LIR

National Internet Registry (NIR)

- NIR is an organization under the umbrella of an RIR with the task of coordinating IP addresses allocations and other Internet resource management functions at a national level within a country or economic unit.
- NIRs operate primarily in the Asia Pacific region, under the authority of APNIC (an RIR for that region)

Regional Internet Registry (RIR)

- An **RIR** is an organization that manages the allocation and registration of Internet number resources within a particular region of the world.
- **Five RIRs (worldwide)**
- African Network Information Centre (**AfriNIC**): Covers Africa region
- American Registry for Internet Numbers(**ARIN**)
 - Covers North America region: United States, Canada, several parts of Caribbean region, Antarctica
- Asia-Pacific Network Information centre (**APNIC**)
 - For Asia/pacific region
- Latin America and Caribbean Network Information Centre (**LACNIC**)
 - Latin America and some Caribbean Islands
- Reseaux IP European Network Coordination Centre (**RIPE NCC**)
 - Europe, Russia, Middle East and Central Asia

Revision ! We are already familiar with , Conversions of $B \rightarrow D$ & $D \rightarrow B$.

Binary:

- All digital electronics use a binary method for communication.
- Binary can be expressed using only two values: 0 or 1.

Converting Binary to Decimal

- Given a binary number, place the number under the chart (right justified).

128	64	32	16	8	4	2	1
1	0	1	1	0	0	1	1

- Add the numbers together to arrive at a final decimal amount.
- $128 + 32 + 16 + 2 + 1 = 179$

Converting Decimal to Binary

- Find the largest number that is equal to or less than the number you are converting to binary. If our example number is 220, the largest number that is equal to or less than 220 is 128. Place a 1 under that space on the chart.
- Next, subtract that number from the original decimal number. Subtracting 128 from 220 gives us 92.
- Repeat this process until we have a subtracted result of 0.

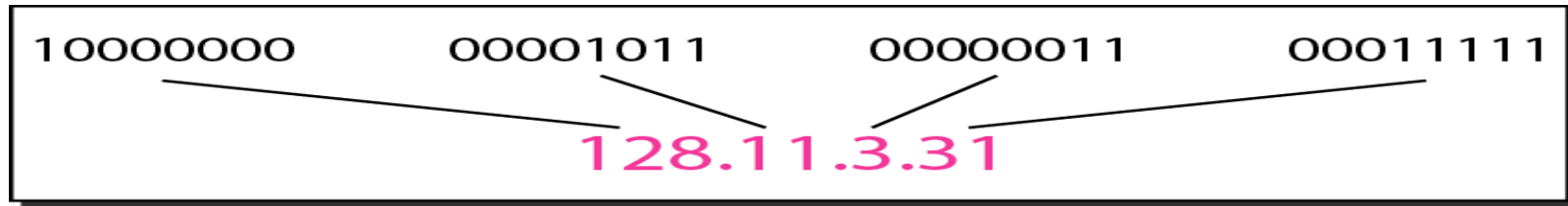
128	64	32	16	8	4	2	1
1	1	0	1	1	1	0	0

IPv4 ADDRESSES

- An *IPv4 address* is a *32-bit* address in length that uniquely and universally defines the connection of a device (for example, a computer or a router) to the Internet.
- The **address space** of IPv4 is **2^{32} or 4,294,967,296**
- To make addressing more humanly manageable, the 32 bits are broken into four 8 bit octets.
- We **separate** the octets by **using a period symbol** – **135.87.252.57**. This is referred to as **dotted decimal notation**.

IPv4 ADDRESSES

Dotted-decimal notation and binary notation for an IPv4 address



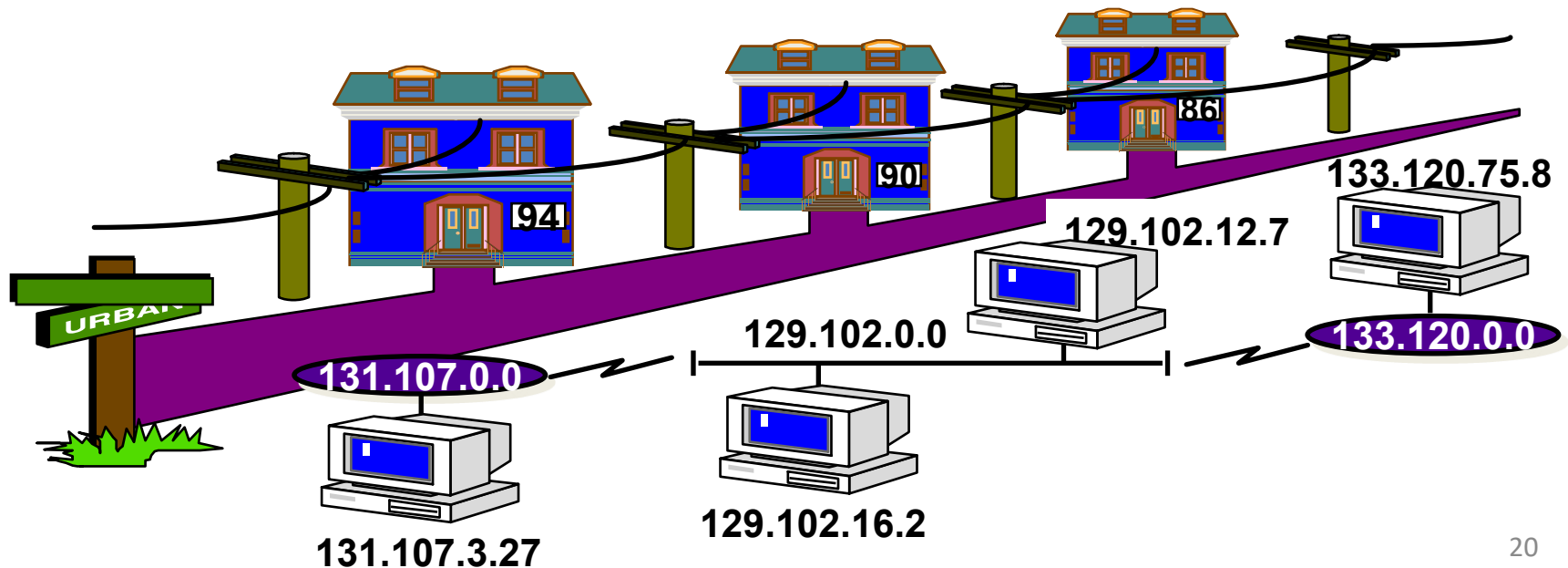
- Errors?*
- a. 111.56.045.78
 - b. 221.34.7.8.20
 - c. 75.45.301.14
 - d. 11100010.23.14.67

IPv4 ADDRESSES Hierarchy

- In any communication system involving delivery, the addressing system is hierarchal.
 - **Postal network** includes country, state, city, street, house number, and the name of the mail recipient.
 - **Telephone network** includes country code, area code, host exchange, and the connection.
- **A 32-bit IP address** is divided into two parts
 - **Prefix** defines the network
 - **Suffix** defines the node (connection of a device to the network)

TCP/IP Host

- A **host** is a device that has a network interface card (NIC) connected to a network.
- If a device has **two** network interfaces, it should be considered **two separate hosts**.
- Each host that is attached to a TCP/IP network must have a **unique TCP/IP address**.



TCP/IP Addresses

- IP Addresses divided into two parts
 - Network ID or **Net ID**
 - Analogous to a street address.
 - **Host ID**
 - Analogous to a house or building number.

Example 1

Change the following IP addresses from binary notation to dotted-decimal notation.

- a. 10000001 00001011 00001011 11101111
- b. 11111001 10011011 11111011 00001111

Solution

We replace each group of 8 bits with its equivalent decimal number and add dots for separation:

- a. 129.11.11.239
- b. 249.155.251.15

Example 2

Change the following IP addresses from dotted-decimal notation to binary notation.

a. 111.56.45.78

b. 75.45.34.78

Solution

We replace each decimal number with its binary equivalent

a. 01101111 00111000 00101101 01001110

b. 01001011 00101101 00100010 01001110

Finding the classes in binary and dotted-decimal notation

In classful addressing, the address space is divided into five classes:

A, B, C, D, and E

	First byte	Second byte	Third byte	Fourth byte
Class A	0			
Class B	10			
Class C	110			
Class D	1110			
Class E	1111			

a. Binary notation

	First byte	Second byte	Third byte	Fourth byte
Class A	0–127			
Class B	128–191			
Class C	192–223			
Class D	224–239			
Class E	240–255			

b. Dotted-decimal notation

Number of blocks and block size in classful IPv4 addressing

Class	Number of Blocks	Block Size	Application
A	128	16,777,216	Unicast
B	16,384	65,536	Unicast
C	2,097,152	256	Unicast
D	1	268,435,456	Multicast
E	1	268,435,456	Reserved

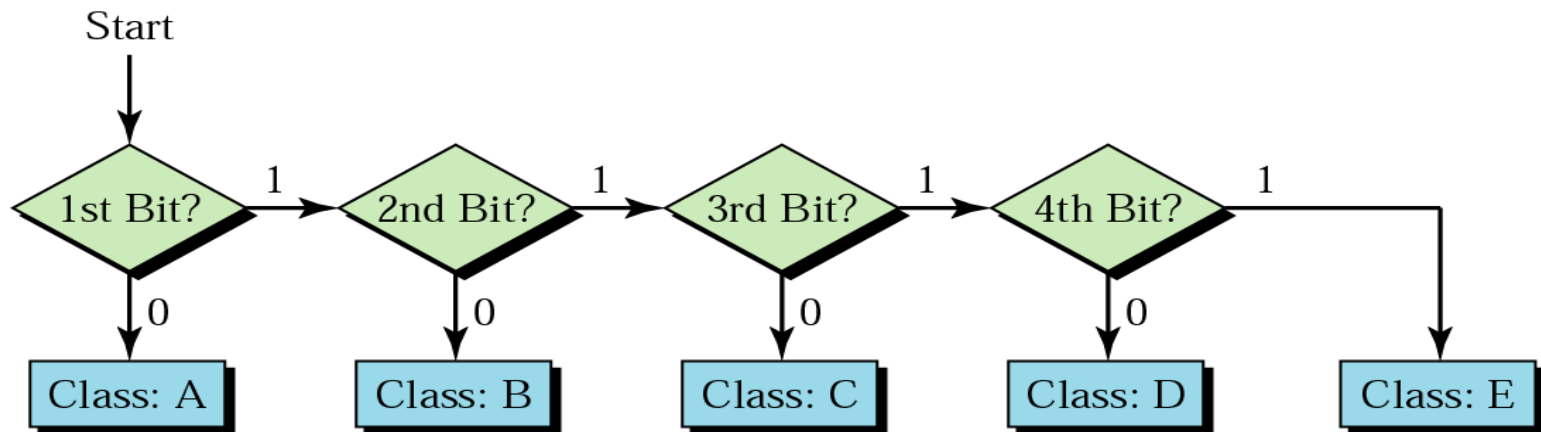
Internet Class-based addresses

- Class A: large number of hosts, few networks
 - 0nnnnnnn hhhhhhhh hhhhhhhh hhhhhhhh
 - 7 network bits (0 and 127 reserved, so 126 networks), 24 host bits (> 16M hosts/net)
 - Initial byte 1-127 (decimal)
- Class B: medium number of hosts and networks
 - 10nnnnnn nnnnnnnn hhhhhhhh hhhhhhhh
 - 16,384 class B networks, 65,534 hosts/network
 - Initial byte 128-191 (decimal)
- Class C: large number of small networks
 - 110nnnnn nnnnnnnn nnnnnnnn hhhhhhhh
 - 2,097,152 networks, 254 hosts/network
 - Initial byte 192-223 (decimal)
- Class D: 224-239 (decimal) Multicast [RFC1112]
- Class E: 240-255 (decimal) Reserved

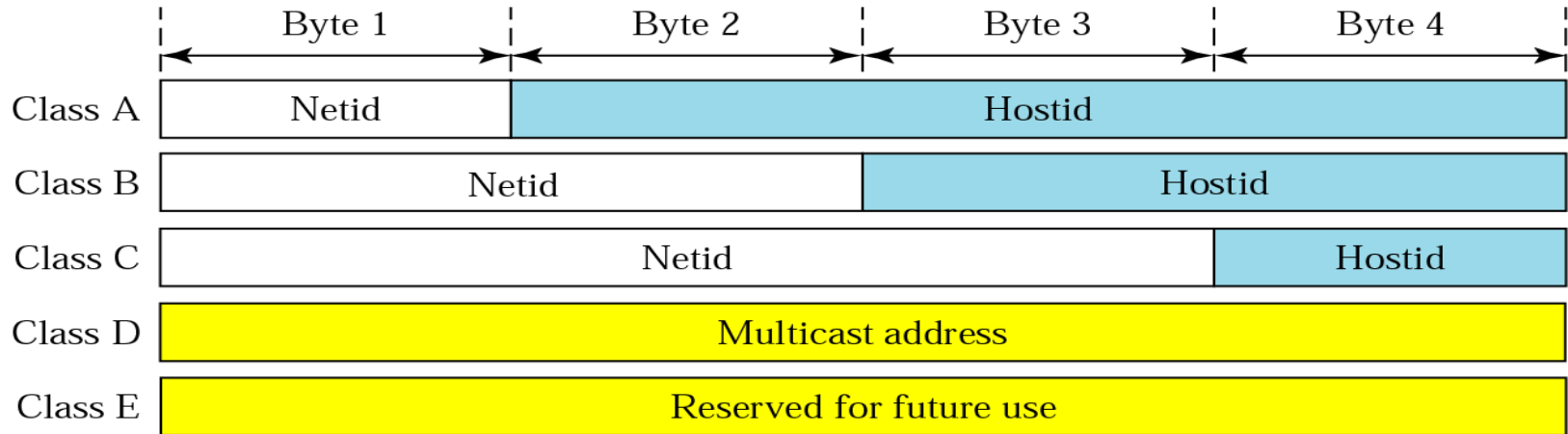
Example

Find the class of each address.

- a. 00000001 00001011 00001011 11101111
- b. 11000001 10000011 00011011 11111111
- c. 11110011 10011011 11111011 00001111
- d. 14.23.120.8
- e. 252.5.15.111
- f. 227.12.14.87



Netid and Hostid



- **Network addresses** cannot be all 0s
- **Hostid: cannot be all 0s**
 - If host portion is all 0s, represents a **network** address.
- **Hostid: cannot be all 1s**
 - If host portion is all 1s, represents **broadcast** address.

Class A Address

- First bit will always be a 0.
- Remaining bits can be either 0s or 1s.
- Range of first octet is 00000000 to 01111111
 - Network addresses cannot be all 0s.
 - 127 is reserved for **loopback testing**
- 126 valid Class A network IDs
 - 1.x.y.z to 126.x.y.z

A **loopback test** is a test in which a signal is sent from a communications device and returned (looped back) to it as a way to determine whether the device is working right or as a way to pin down a failing node in a network

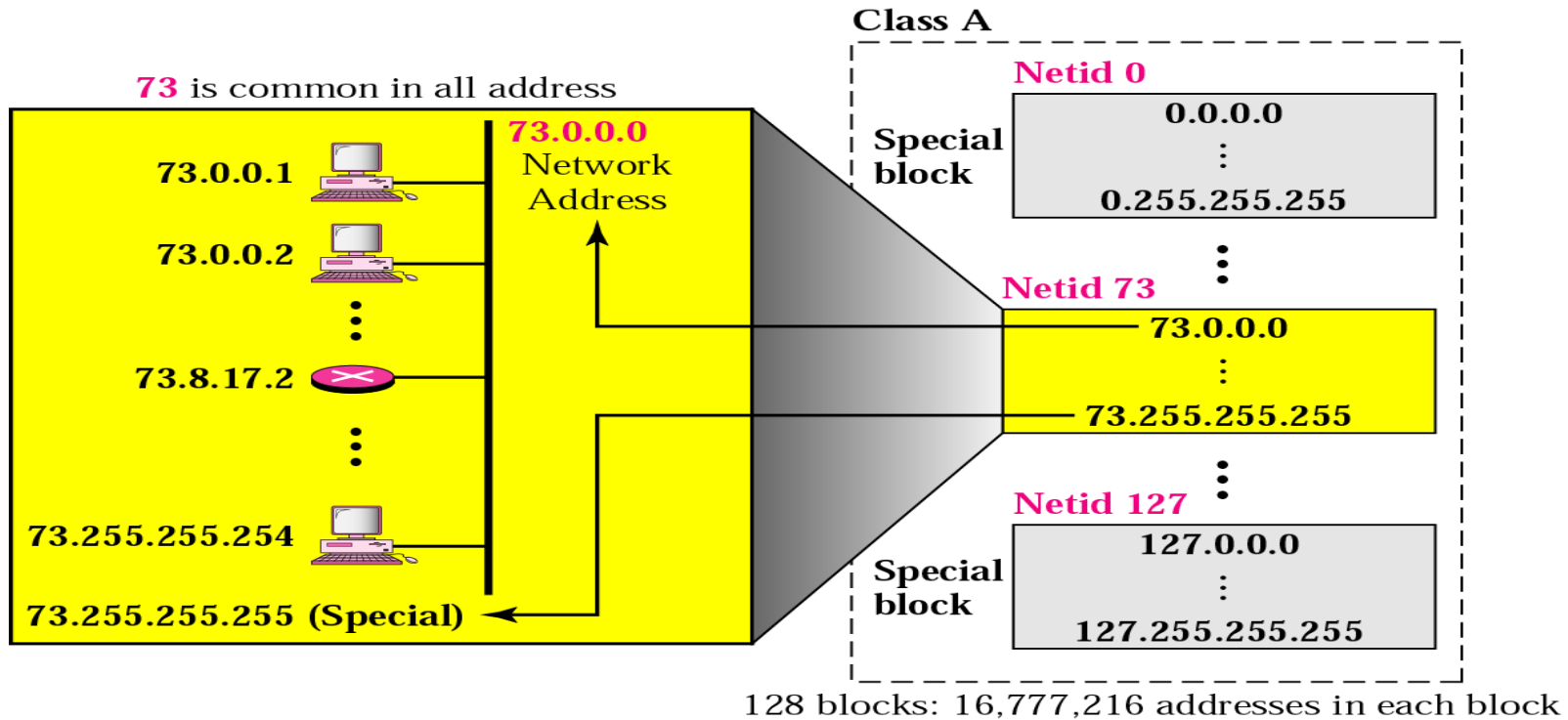
Ref: <http://searchnetworking.techtarget.com/definition/loopback-test>

Class A Address

The address range from **0.0.0.0** through **0.255.255.255** should not be considered part of the normal Class A range. 0.x.x.x addresses serve no particular function in IP, but nodes attempting to use them will be unable to communicate properly on the Internet.

For details of special Use IPv4 addresses:
<http://tools.ietf.org/html/rfc5735>

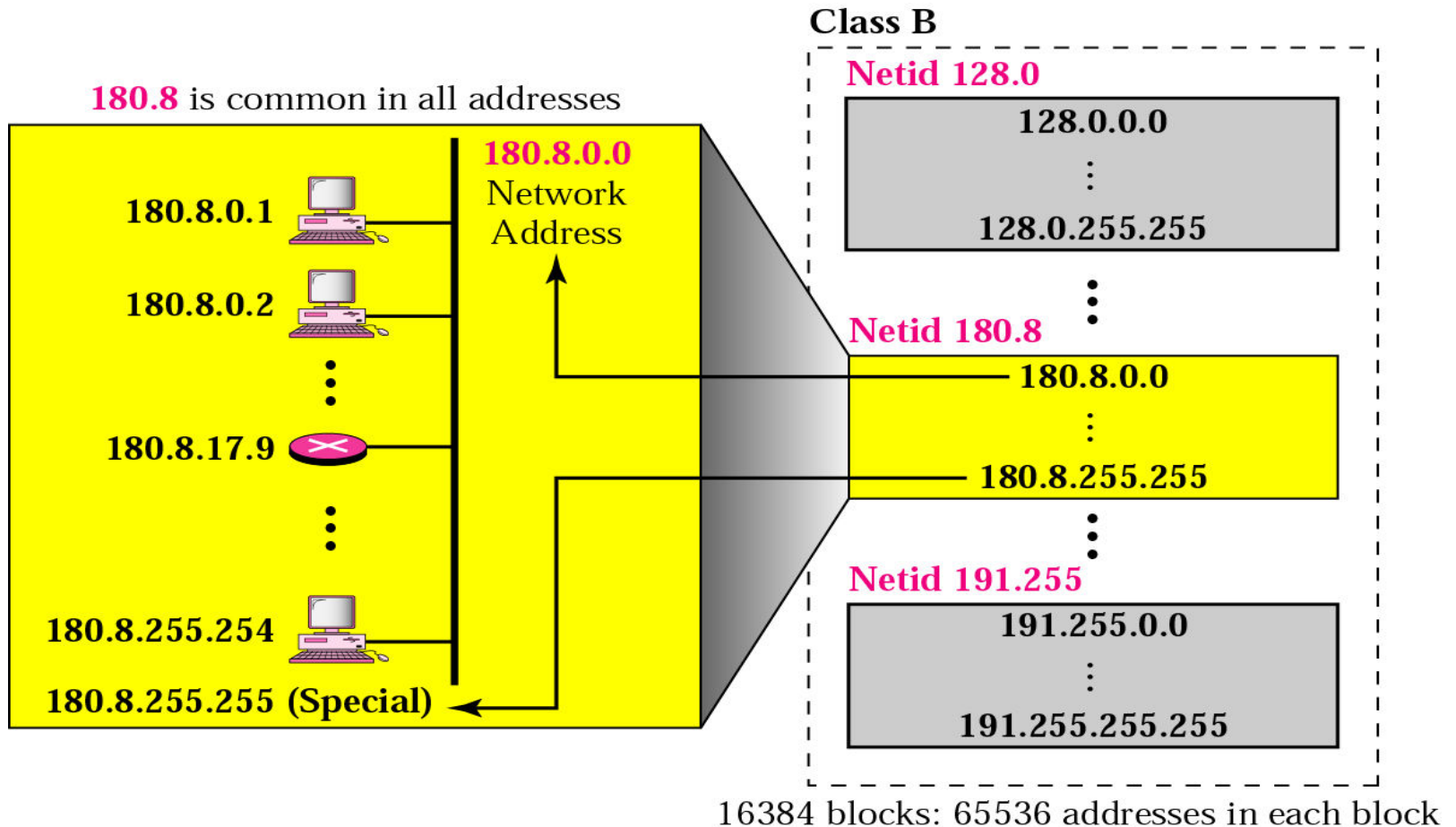
Blocks in class A



Class B Address

- First two bits will always be a 10.
- Remaining bits can be either 0s or 1s.
- Range of first octet is 10000000 to 10111111
- Range of networks 128.0.y.z to 191.255.y.z
- 16,384 valid Class B network IDs.

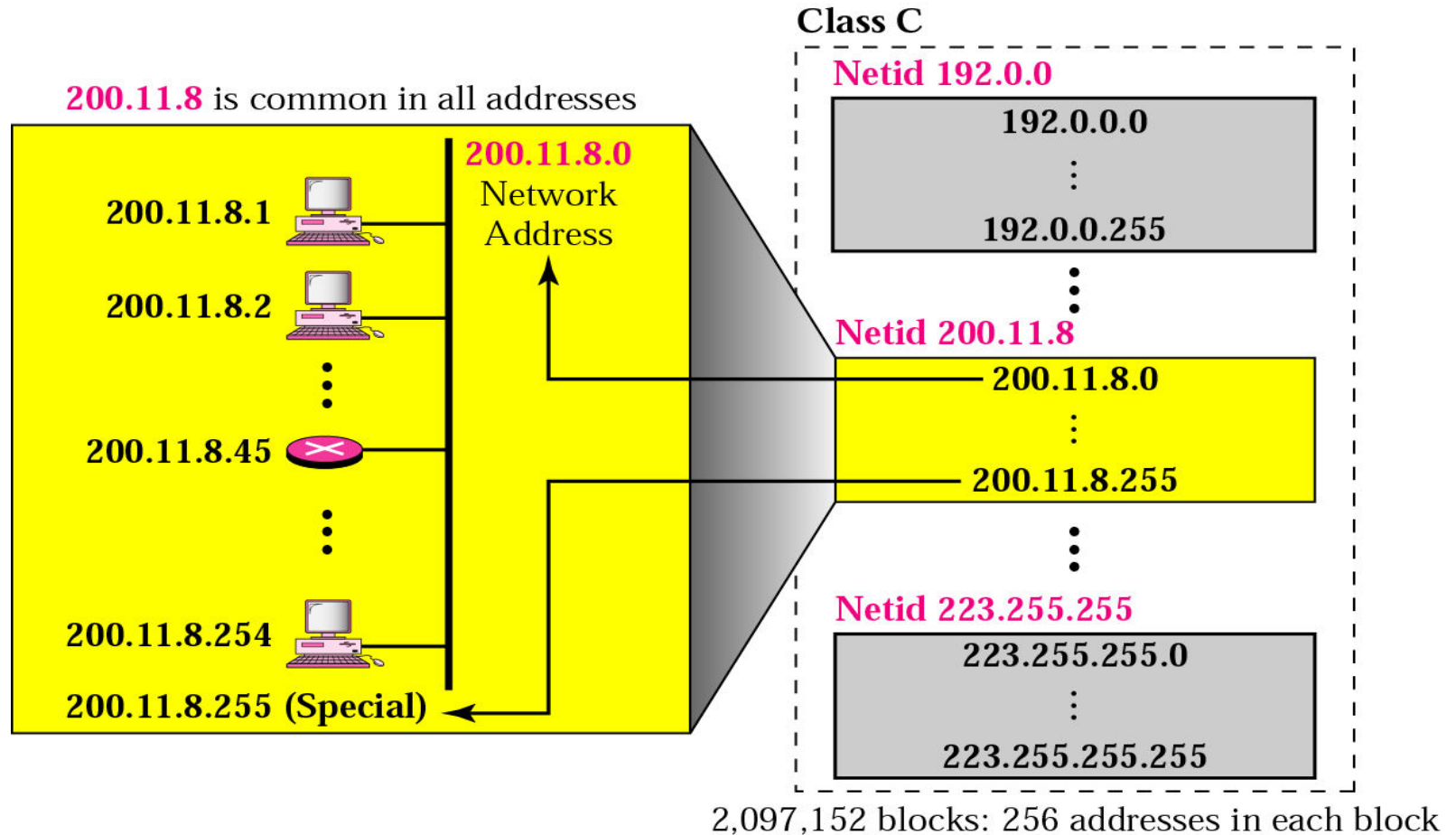
Blocks in class B



Class C Address

- First three bits will always be a 110.
- Remaining bits can be either 0s or 1s.
- Range of first octet is 11000000 to 11011111
- Range of class C networks is 192.0.0.z to 223.255.255.z.
- 2,097,152 valid Class C network IDs.

Blocks in class C



Class D Address

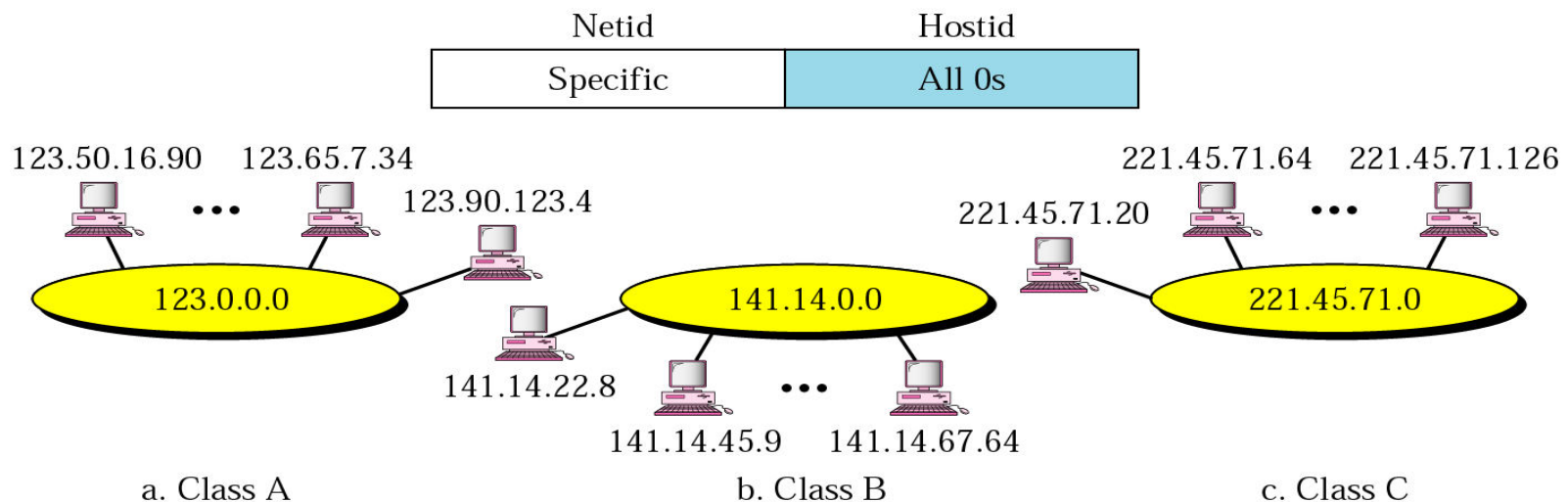
- First octet in binary is defined as 1110xxxx, replacing x's with whatever we wish.
- Range of Class D addresses is from 224.x.y.z to 239.x.y.z.
- **Used for multicasting** – method of sending a single packet to multiple hosts.

Class E Address

- First octet is 1111xxxx, replacing x's with whatever we wish.
- Address ranges from 240.x.y.z to 255.x.y.z.
- **Experimental address range** that is not used in actual networks.

Network Address

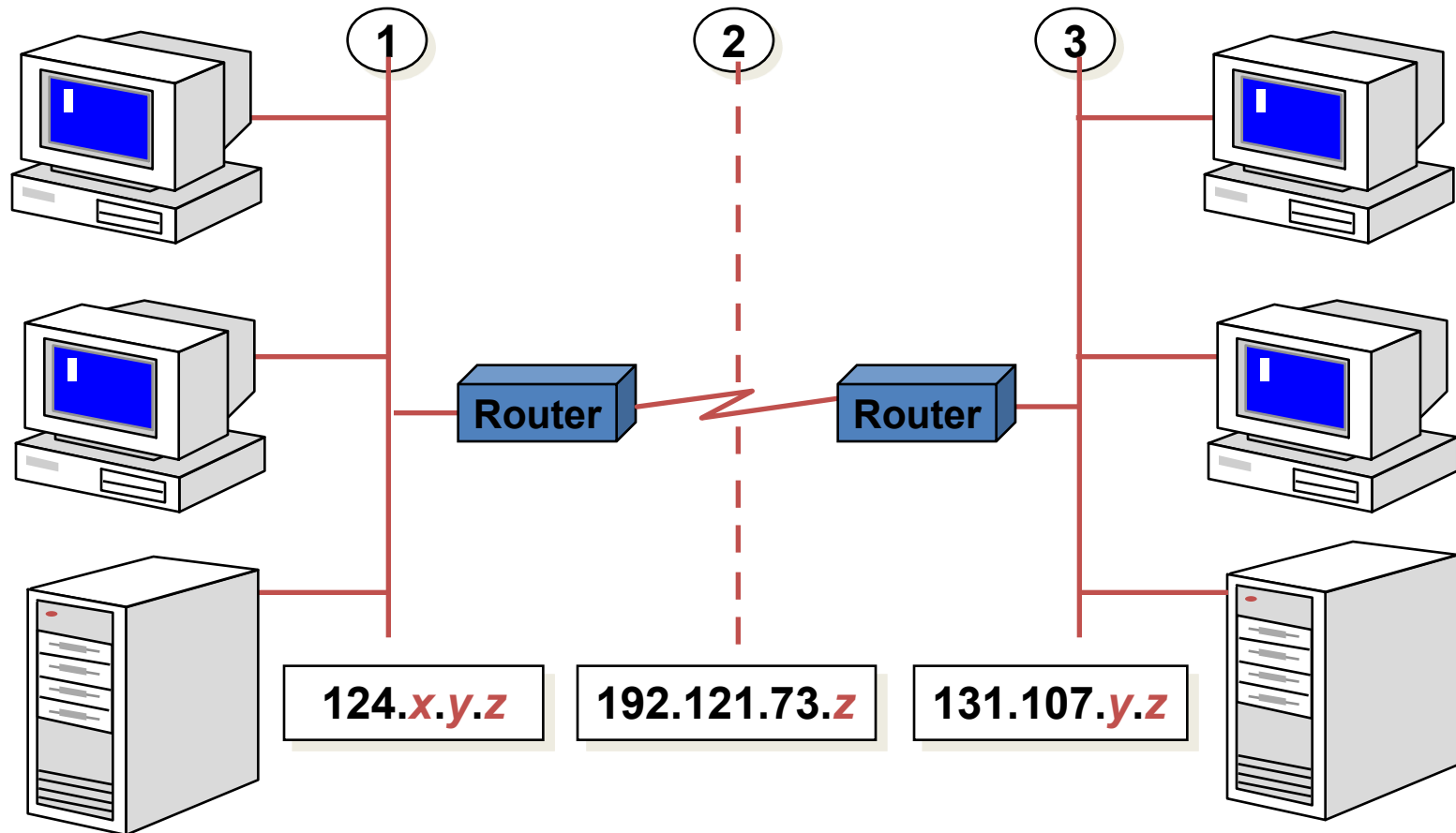
- A **network address** is different from a **netid**.
 - A network address has **both netid and hostid**, with **0s** for the **hostid**
 - The **first address** is called the **network address** and defines the **organization network**.
 - It defines the organization itself to the rest of the world.
 - The **organization network** is **connected** to the **Internet** via a **router**.
- The router has **two addresses**. One belongs to the **granted block**; the other belongs to the network that is **at the other side of the router**.



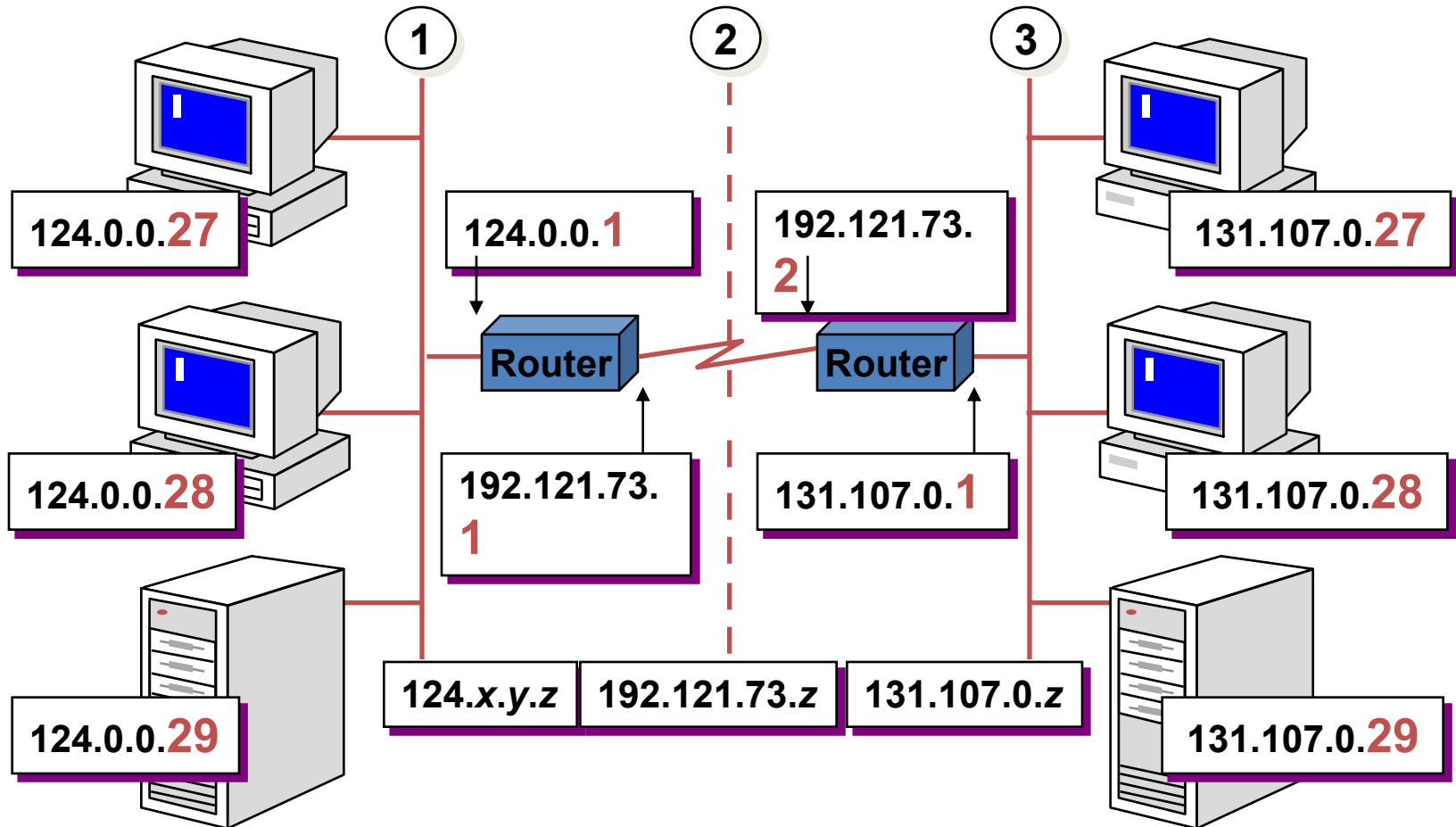
Examples: Network Address

- Given the address **23.56.7.91**, find the **network address**.
 - The class is **A**. Only the first byte defines the **netid**. We can find the network address by replacing the hostid bytes (56.7.91) with 0s. Therefore, the network address is **23.0.0.0**.
- Given the address **132.6.17.85**, find the **network address**
 - The class is **B**. The first 2 bytes defines the netid. We can find the network address by replacing the hostid bytes (17.85) with 0s. Therefore, the network address is **132.6.0.0**.

Assigning Network IDs



Assigning Host IDs



Address Class Summary

[[

	Number of Networks	Number of Hosts per Network	Range of Network IDs (First Octet)
Class A	126	16,777,214	1 – 126
Class B	16,384	65,534	128 – 191
Class C	2,097,152	254	192 – 223

Addressing Guidelines

- Network ID cannot be 0 (all Bits set to 0)
 - Serve no particular purpose in IP
- Network ID cannot be 127
 - 127 is reserved for loopback functions
- Host ID cannot be 255 (All Bits Set to 1)
 - 255 is a broadcast address
- Host ID cannot be 0 (All Bits Set to 0)
 - 0 means “this network only”
 - Example: 145.20.0.0 refers to Class B network
145.20.0.0
- Host ID Must Be unique to the Network

Reserved, Private addresses

Private address block:

Class A: 10.0.0.0 to 10.255.255.255

Class B: 169.254.0.0 to 169.254.255.255
and 172.16.0.0 to 172.31.255.255

Class C: 192.168.0.0 to 192.168.255.255

Reserved:

Class A: 0.0.0.0 to 0.255.255.255 & 127.0.0.0 to 127.255.255.255 (LB)

Class B: 128.0.0.0 to 128.0.255.255 & 191.255.0.0 to 191.255.255.255

Class C: 192.0.0.0 to 192.0.0.255 & 223.255.255.0 to 223.255.255.255